



Paper ID : 261148

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Subject Code: MPH202T

Roll No:

MPHARM

(SEM. II) THEORY EXAMINATION 2025-26

ADVANCED BIOPHARMACEUTICS & PHARMACOKINETICS

TIME: 3 HRS

M.MARKS: 75

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief.

2x10 = 20

- (a) Enumerate the factors affecting absorption of drugs.
- (b) Elaborate the abbreviation IVIVC.
- (c) What are tight junction complex?
- (d) Enlist factors affecting the dissolution rate.
- (e) Mention the rate-limiting steps in drug absorption.
- (f) What are the alternative methods of dissolution testing?
- (g) Mention the purpose of bioavailability studies.
- (h) Differentiate between pharmacokinetics and pharmacodynamics.
- (i) Give the BCS classification of drugs.
- (j) How will you define drug interactions?

SECTION B

2. Attempt any two of the following:

2 x 10 = 20

- (a) Discuss in detail the pH partition theory of absorption. Give its limitations.
- (b) Enumerate and explain various physicochemical factors affecting absorption of drug.
- (c) What do you understand by compendial methods of dissolution? Explain any one in detail.

SECTION C

3. Attempt any five part of the following:

7 x 5 = 35

- (a) Determine the absorption rate constant using Wagner Nelson method.
- (b) What are bioequivalence studies? Elaborate the study designs and crossover study designs.
- (c) Explain drug dissolution rate using Noyes Whitney equation signifying various factors affecting the same.
- (d) Describe in detail the different direct methods for the assessment of bioavailability.
- (e) Discuss the different levels of IVIVC.
- (f) Compare pharmacokinetics and pharmacodynamics. Write a detailed note emphasizing the drug interactions.
- (g) Elaborate the pharmacokinetic applications of modified release drug products.